SHIVAM KOTAK

Los Angeles, CA-90007 | (213) 414-9656 | shivamkotak@gmail.com | LinkedIn: shivamkotak | t3ds.github.io

EDUCATION

University of Southern California, Los Angeles, CA **January 2022-December 2023**
Master of Science in Computer Science **CGPA:3.8**
Relevant Coursework**:** Applied NLP, Foundations of AI, Machine Learning, Web Technologies, Database Systems, Information Retrieval

University of Mumbai, Mumbai, India **August 2017-July 2021**
Bachelor of Engineering in Computer Engineering **CGPA:9.1/10**

EXPERIENCE

your.MOV, Los Angeles, USA **March 2024-Present**
Software Engineer

- Joined at inception stage and played a **key role in building a video editing platform from the ground up**
- **Designed** and **iterated** on the **React front-end**, **shaping the product vision** and **coordinating with contractors** to guide implementation.
- **Engineered critical Django backend** components, ensuring **efficiency** and **reliability** for **performance-sensitive workflows**.
- Developed **C++ rendering logic** powering the video pipeline, including core code paths that interface with the renderer.
- Applied **LLM systems** and **prompt engineering** to enable automated editing workflows.
- **Contributed** to **AWS infrastructure (Lambda, ECS, S3, CI/CD pipelines),** supporting **scalability** and **deployment reliability**.
- **Recognized** by leadership as a **critical team member** for **adaptability** and ability to contribute across **frontend**, **backend**, **machine learning**, and **systems programming**.

CCC Intelligent Solutions, Chicago, USA
Data Science Intern **August 2023-December 2023**

- **Engineered prompts** for **IDEFICS** and **Cheetah (Llama Based) visual language models** for the task of **Visual-Question Answering** for damage summarization, helping clients understand claim information in seconds instead of minutes.
- **Deployed into production** a pipeline to remove PII in both **visual** and **textual** data with help of **AWS Step Functions, Lambdas, CloudFormation, EventBridge**., scheduled to be nightly batched. The pipeline enhances the feature store being used by the Data Science department to create inbound subrogation models and is expected to **process 100000 demands per week**.
- **Created** a package of metrics to **evaluate generated text with ground truths**. The package provides **similarity-based metrics** like **BERT embeddings** as well as metrics relying on **distance measures like fuzzy edit distance** and will be used to **evaluate outputs of all generative language models being developed** by the company.
- Worked on a **machine learning model pipeline using CLIPSeg** for evidence verification in subrogation claims to help increase success rates in subrogation disputes.
- Wrote **unit tests** and **regression tests** with benchmarks using **pytest** and **pytest_benchmark** to ensure smooth functionality of code.
- success rates in subrogation disputes.

TECHNICAL SKILLS

- **Programming Languages:** Python, C, C++, Java |**Web Development Stack:** HTML, CSS, Vanilla JS, React JS, Angular, Typescript, PHP, Flask, AWS (Step Functions, Lambdas, EventBridge, CloudFormation, Sagemaker, S3, EC2), GCP, Node.js, Android, Express.js, Bootstrap, Material UI | **Database:** SQL, Oracle SQL, PostgreSQL, MongoDB, DynamoDB
- **Machine Learning/Data Science:** PyTorch, TensorFlow, Keras, Scikit-learn, XGBoost, SpaCy, NLTK, OpenCV, AWS (Sagemaker, S3, EC2, Lambda), MLFlow, Scipy, Numpy, Pandas, Transformers, Gen AI, Matplotlib, Seaborn, Optuna, HuggingFace
- **Frameworks and Tools:** JavaFX, PowerBi, Git, Docker, Hadoop, Bash, Linux, Gitlab, ffmpeg, Mediapipe

ACADEMIC PROJECTS

Stock Price Prediction Dashboard – Link to Repo – Python, TensorFlow, Flask, React.js

- Predicted end-of-day share price for a company on the National Stock Exchange. Leveraged the **Keras** library in Python by employing a **CNN-LSTM** neural network. Employed **React.js**, **Material UI** and **chart.js** for the front end.
- Collated historical data using a python library and evaluated data based on technical indicators such as **RSI, MACD** etc
- Trained a prediction model based on past 4-5 years data. Predicted values within a variation of approximately **10-15%** of the actual price.

Manga Downloader – Link to Repo – Java, JavaFX, Web Scraping, HTML Scraping, Selenium, BeautifulSoup4

- Developed an application that takes input as a name of a Japanese comic and then lists and downloads chapters of a comic
- Coded in **Java**, leveraging **JavaFX** for the UI. It works by parsing the HTML of a source website (mangadex.org) for identifying, listing, and downloading chapters. The pages were scraped by creating a selenium session to read the HTML of dynamically loaded web pages.

Football Commentary Guided Player Rating Prediction – Link to Repo – Python, PyTorch

- Explored the use of **BERT** and **XLNet** to encode commentaries to test on different models (Both Classical and Deep Models) with Negative Log Likelihood loss and MSE loss. The best result loss of **0.35** was achieved with **XLNet** and an **LSTM** model.
- Gathered data by collating from multiple datasets and API. Final processed dataset can be found on Kaggle at: Football Player Rating, Stats and Event Commentary

LEADERSHIP AND INVOLVEMENT

- As part of an early state startup, took initiative for features and directions of product as well as managed contracted employees to ensure effective work. Also contributed to write up proposals for investments.